

Get started with data engineering on Azure

Learning Objectives

After completing this module, you will be able to:

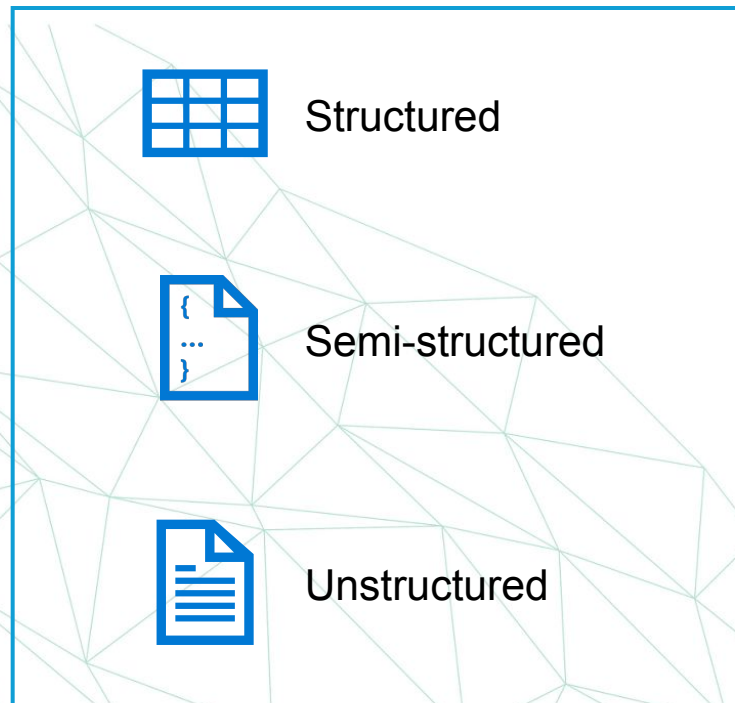
- 1 Introduction to data engineering on Azure
- 2 Introduction to Azure Data Lake Storage Gen2
- 3 Introduction to Azure Synapse Analytics

Introduction to data engineering on Azure

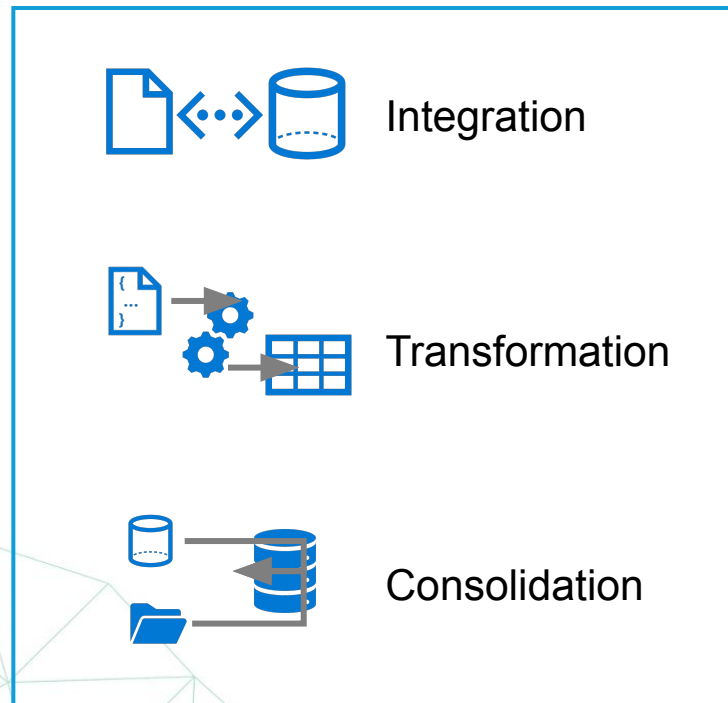
What is data engineering?

Data engineers work with multiple types of data to perform a variety of data operations using a range of tools and scripting languages

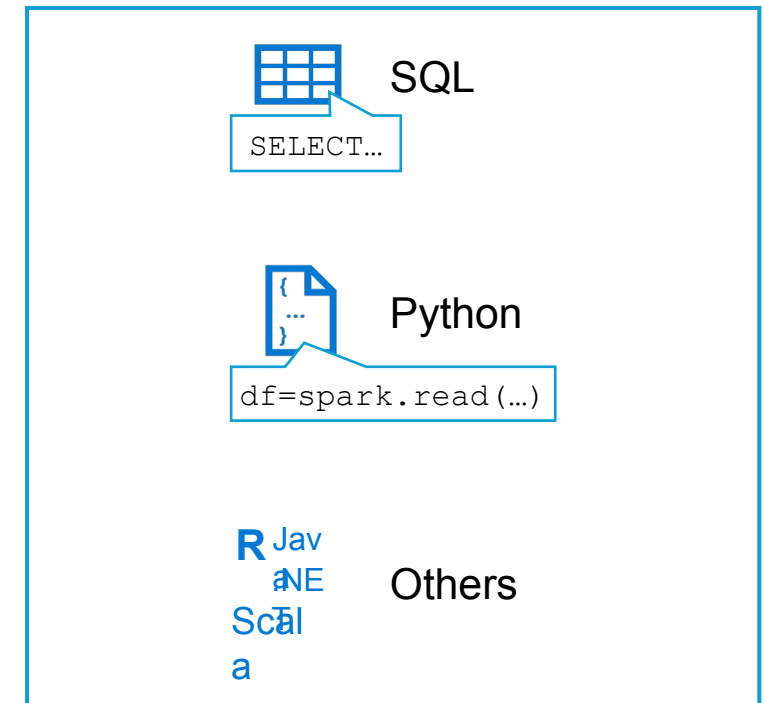
Types of data



Data operations

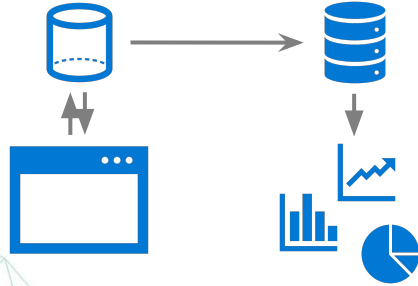


Languages



Important data engineering concepts

Operational and analytical data



Operational: Transactional data used by applications
Analytical: Optimized for analysis and reporting

Streaming data



Perpetual, real-time data feeds

Data pipeline



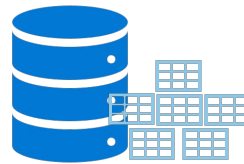
Orchestrated activities to transfer and transform data.
Used to implement *extract, transform, and load* (ETL) or *extract, load, and transform* (ELT) operations.

Data Lake



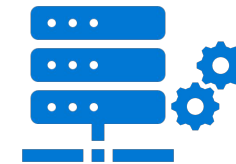
Analytical data stored in files
Distributed storage for massive scalability

Data Warehouse



Analytical data stored in a relational database
Typically modeled as a star schema to optimize summary analysis

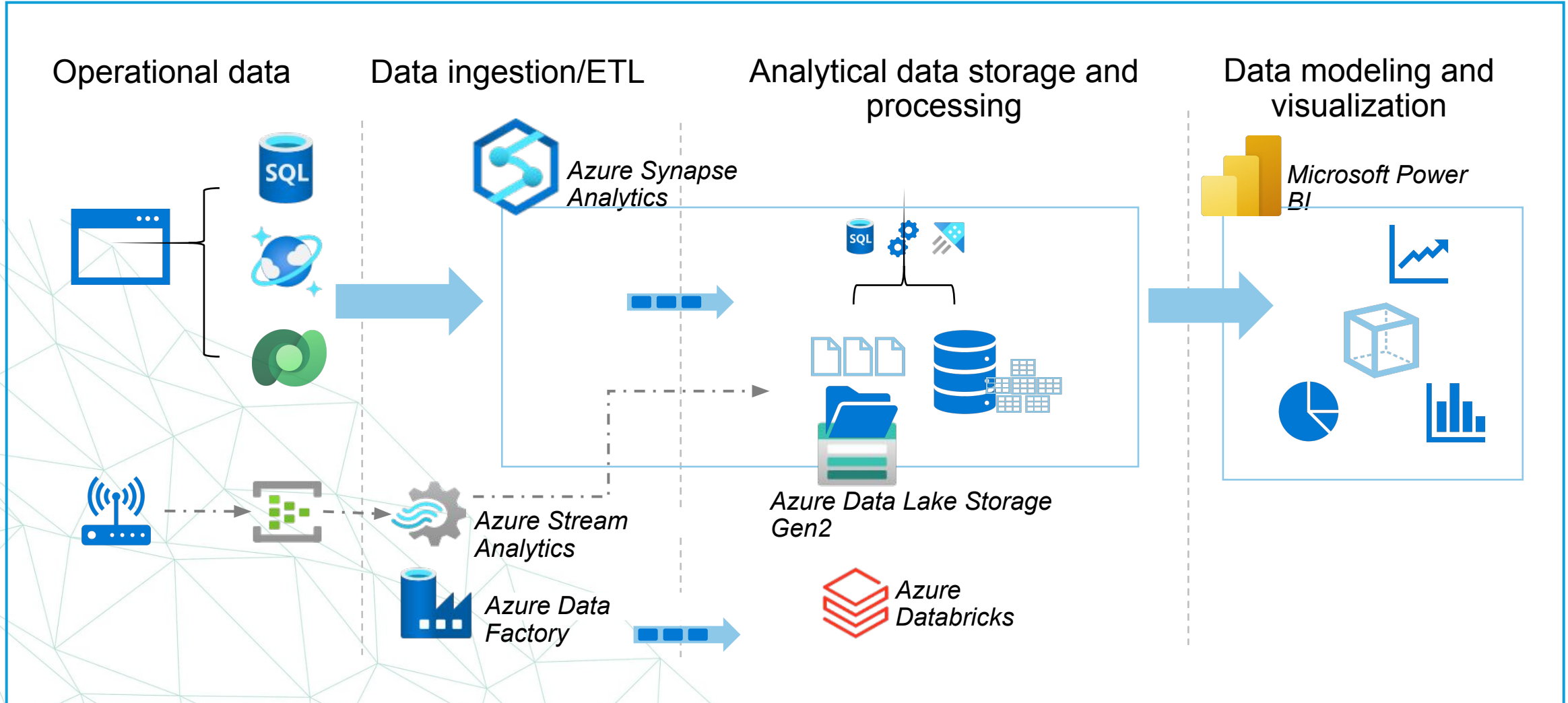
Apache Spark



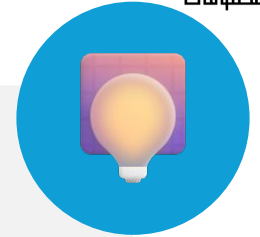
Open-source engine for distributed data processing

	Data lake	Data warehouse
Type	Structured, semi-structured, unstructured	Structured
	Relational, non-relational	Relational
Schema	Schema on read	Schema on write
Format	Raw, unfiltered	Processed, vetted
Sources	Big data, IoT, social media, streaming data	Application, business, transactional data, batch reporting
Scalability	Easy to scale at a low cost	Difficult and expensive to scale
Users	Data scientists, data engineers	Data warehouse professionals, business analysts
Use cases	Machine learning, predictive analytics, real-time analytics	Core reporting, BI

Data engineering in Azure



Knowledge check



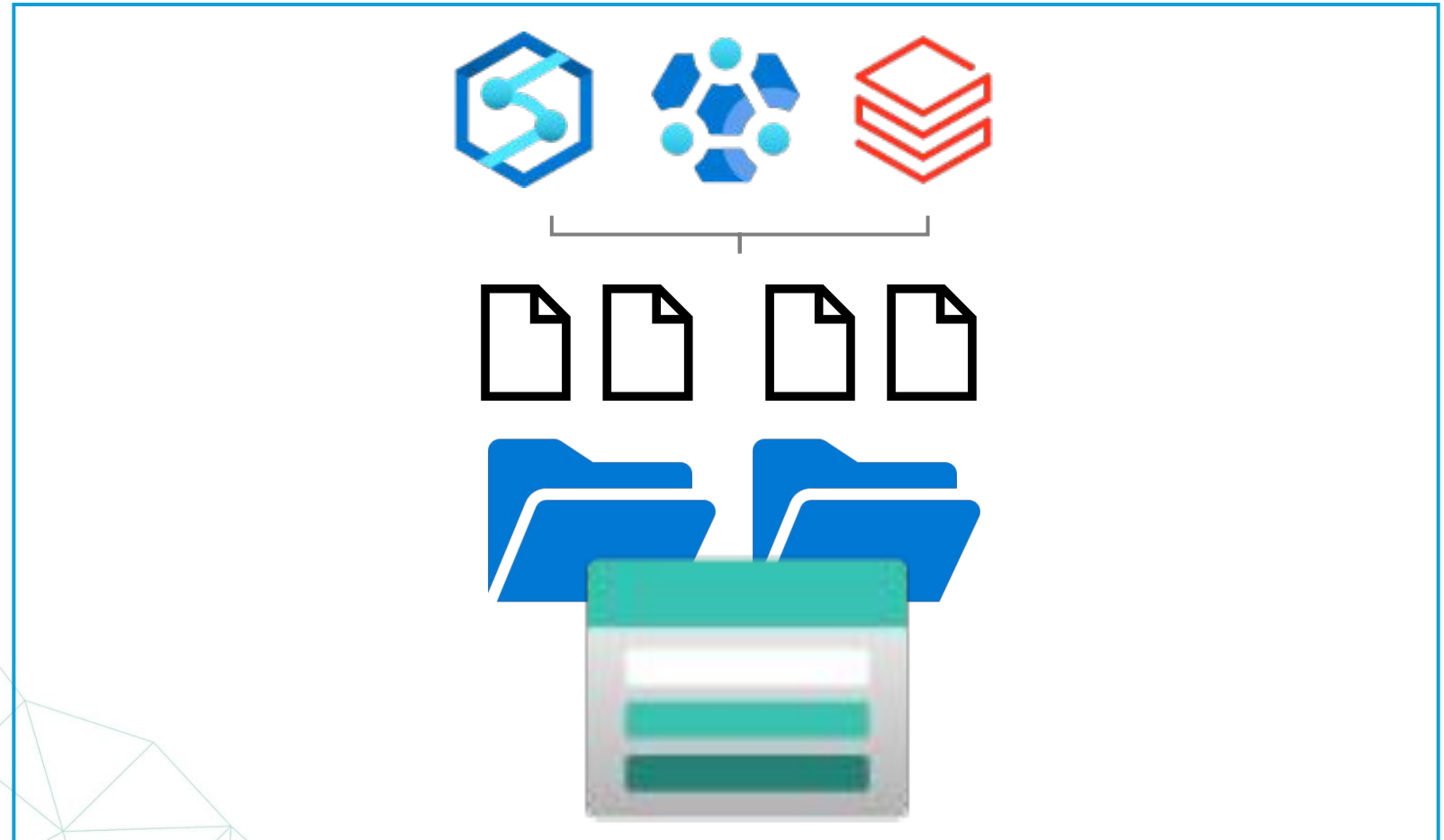
- 1 Data in a relational database table is...
 - ☒ Structured
 - ☐ Semi-structured
 - ☐ Unstructured
- 2 In a data lake, data is stored in...
 - ☐ Relational tables
 - ☒ Files
 - ☐ A single JSON document
- 3 Which of the following Azure services provides capabilities for running data pipelines AND managing analytical data in a data lake or relational data warehouse?
 - ☐ Azure Stream Analytics
 - ☒ Azure Synapse Analytics
 - ☐ Azure Databricks

Introduction to Azure Data Lake Storage Gen2

Understand Azure Data Lake Storage Gen2

Distributed cloud storage for data lakes

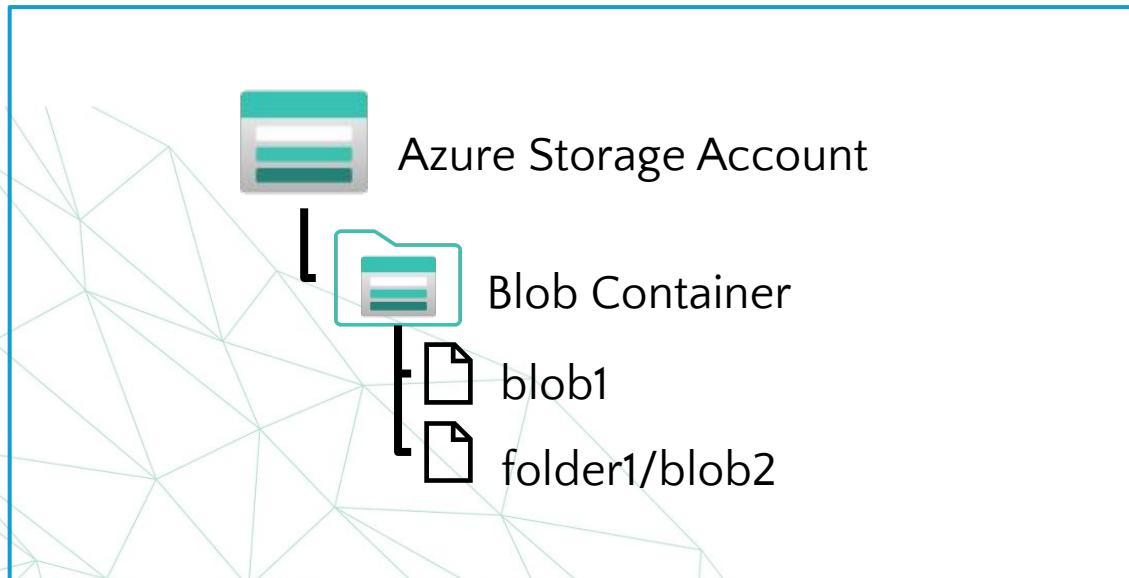
- HDFS-compatibility – Common file system for Hadoop, Spark, and others
- Flexible security through folder and file level permissions
- Built on Azure Storage:
 - High performance and scalability
 - Data redundancy through built-in replication



Azure Data Lake Storage Gen 2 vs Azure Blob Storage

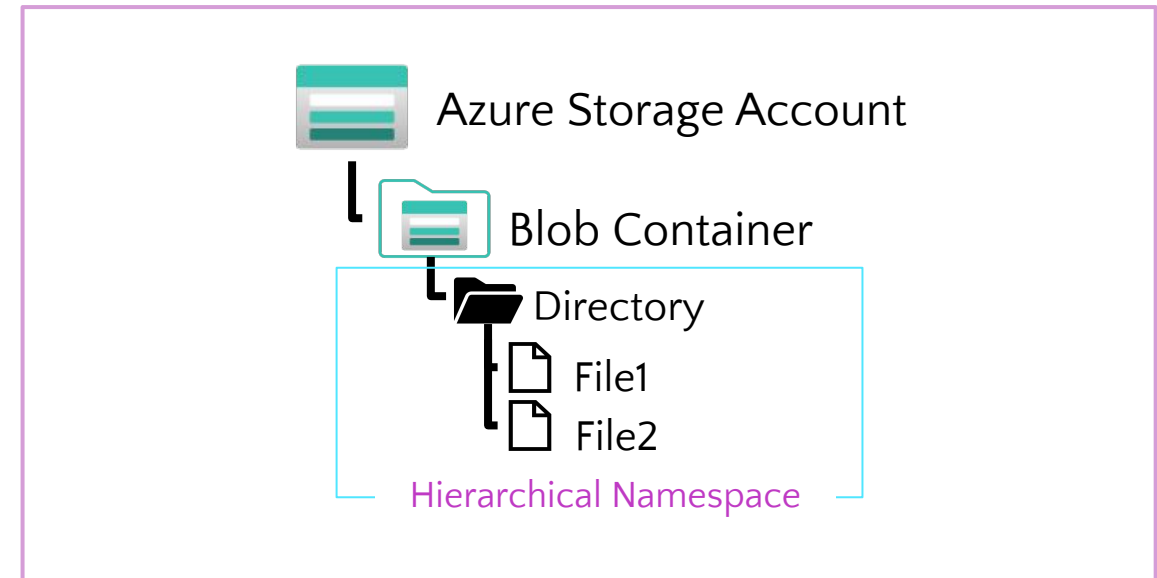
Enable *Hierarchical Namespace* in a blob container to use Azure Data Lake Storage Gen2

Azure Blob Storage



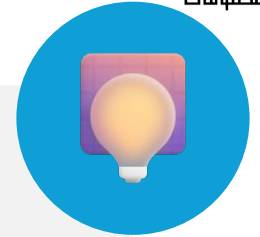
Blobs can be organized in virtual directories, but each path is considered a single blob in a flat namespace – Folder level operations are not supported

Azure Data Lake Storage Gen2



File system includes directories and files, and is compatible with large scale data analytics systems like Hadoop, Databricks, and Azure Synapse Analytics

Knowledge check



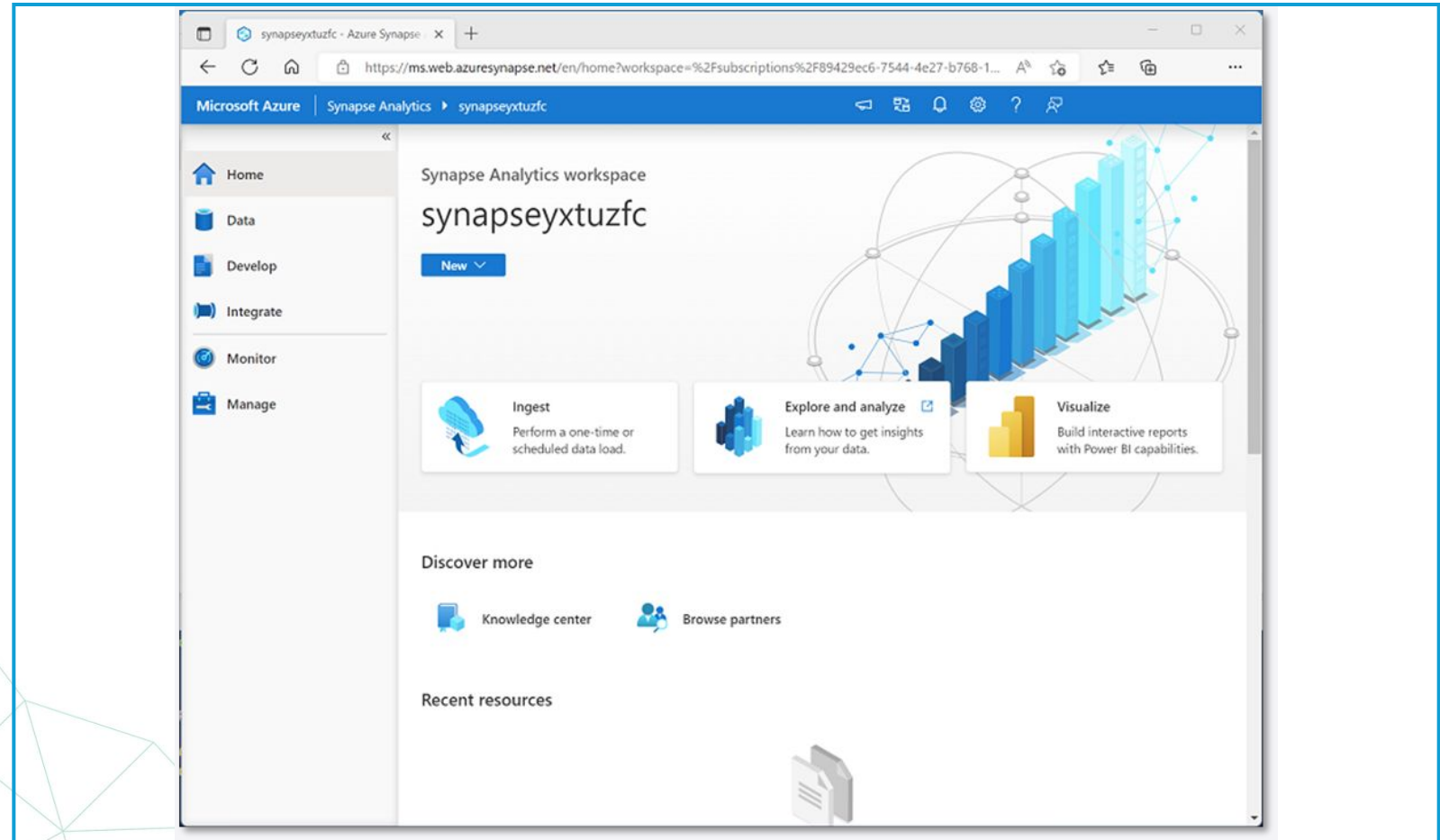
- 1 Azure Data Lake Storage Gen2 stores data in...
 - ☐ A document database hosted in Azure Cosmos DB
 - ☒ An HDFS-compatible file system hosted in Azure Storage
 - ☐ A relational data warehouse hosted in Azure Synapse Analytics
- 2 What option must you enable to use Azure Data Lake Storage Gen2?
 - ☐ Global replication
 - ☐ Data encryption
 - ☒ Hierarchical namespace

Introduction to Azure Synapse Analytics

What is Azure Synapse Analytics?

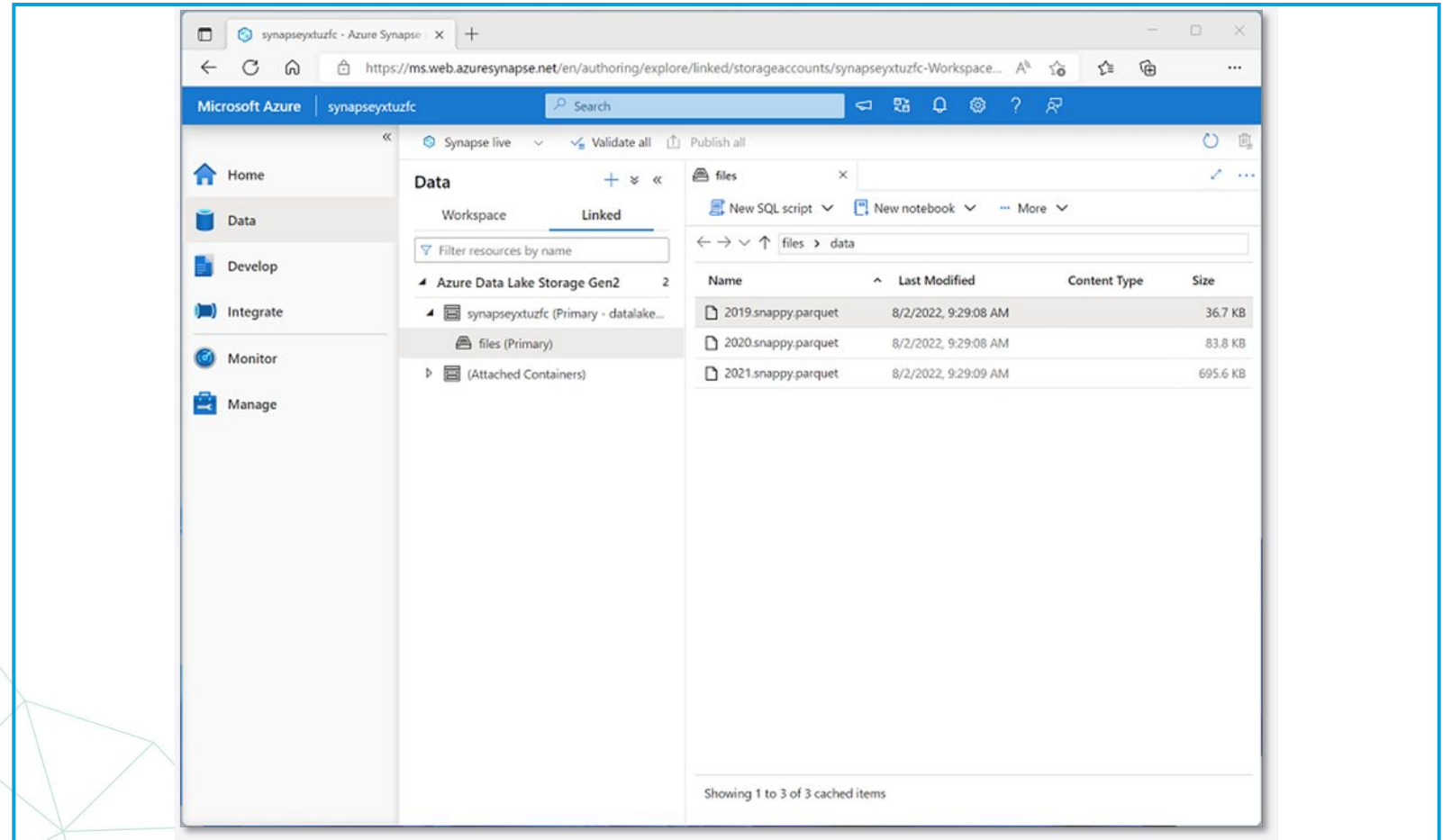
Cloud platform for data analytics

- Large-scale data warehousing
- Advanced analytics
- Data exploration and discovery
- Real time analytics
- Data integration
- Integrated analytics



Work with files in a data lake

- Connect to data lake storage using *linked services*
- Every Azure Synapse Analytics workspace has a default data lake

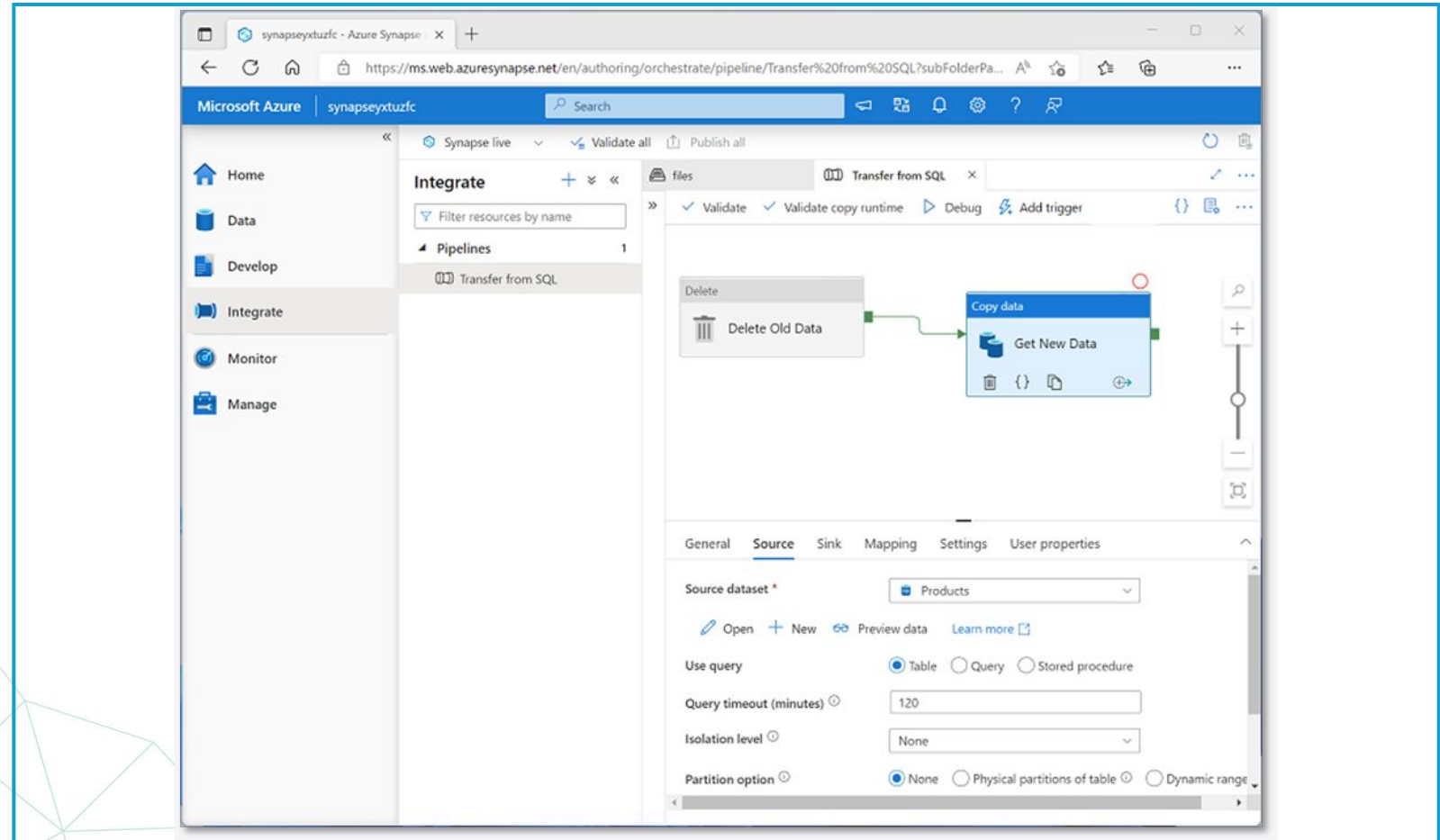


The screenshot displays the Azure Synapse Analytics portal interface. On the left, a navigation pane shows options like Home, Data, Develop, Integrate, Monitor, and Manage. The main area is titled 'Data' and includes a 'Workspace' tab and a 'Linked' tab. Under the 'Linked' tab, there's a search bar and a list of resources. One resource, 'files (Primary)', is expanded, showing a table of files. The table has columns for Name, Last Modified, Content Type, and Size. It lists three files: '2019.snappy.parquet' (36.7 KB), '2020.snappy.parquet' (83.8 KB), and '2021.snappy.parquet' (695.6 KB). The bottom of the table indicates 'Showing 1 to 3 of 3 cached items'.

Name	Last Modified	Content Type	Size
2019.snappy.parquet	8/2/2022, 9:29:08 AM		36.7 KB
2020.snappy.parquet	8/2/2022, 9:29:08 AM		83.8 KB
2021.snappy.parquet	8/2/2022, 9:29:09 AM		695.6 KB

Ingest and transform data with pipelines

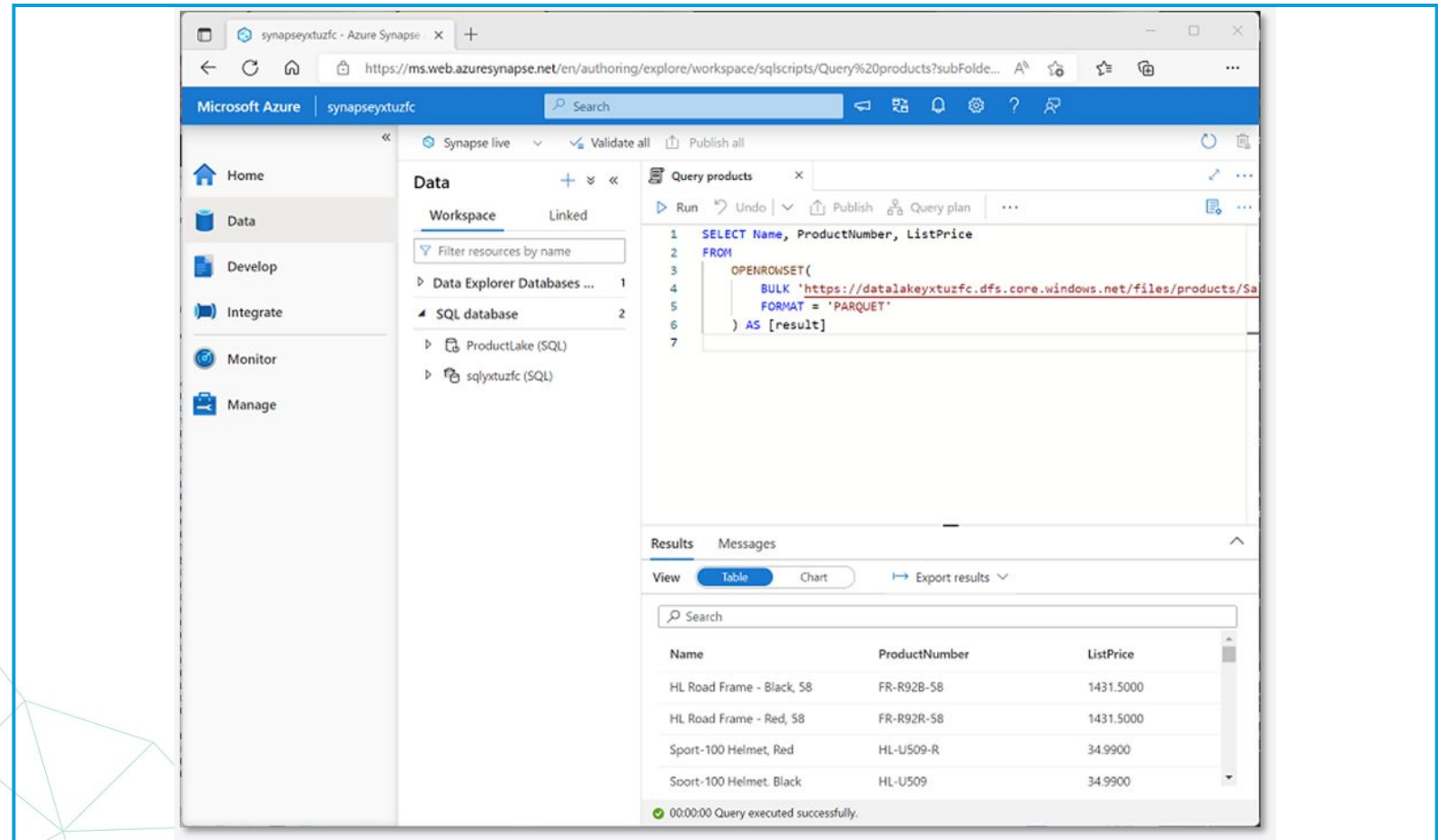
- Native pipeline functionality built on Azure Data Factory
- Orchestrate activities to ingest, transform, and load data
- Integrate with other data services



Query and manipulate data with SQL

SQL Server based pools for scalable relational data processing:

- Built-in *serverless* SQL pool for data exploration and analysis of files in the data lake
- Custom *dedicated* SQL pools to host large-scale relational data warehouses



The screenshot displays the Microsoft Azure Synapse Studio interface. The left sidebar shows navigation options: Home, Data, Develop, Integrate, Monitor, and Manage. The main workspace is divided into two panes. The left pane, titled 'Data', shows a tree view of resources including 'Data Explorer Databases ...', 'SQL database', 'ProductLake (SQL)', and 'sqlxtzfc (SQL)'. The right pane, titled 'Query products', shows a SQL query being executed. The query is as follows:

```
1 SELECT Name, ProductNumber, ListPrice
2 FROM
3 OPENROWSET(
4     BULK 'https://datalakeyxtzfc.dfs.core.windows.net/files/products/Sa
5     FORMAT = 'PARQUET'
6 ) AS [result]
7
```

Below the query editor, the 'Results' pane shows the output of the query in a table view. The table has three columns: Name, ProductNumber, and ListPrice. The results are as follows:

Name	ProductNumber	ListPrice
HL Road Frame - Black, 58	FR-R92B-58	1431.5000
HL Road Frame - Red, 58	FR-R92R-58	1431.5000
Sport-100 Helmet, Red	HL-U509-R	34.9900
Sport-100 Helmet, Black	HL-U509	34.9900

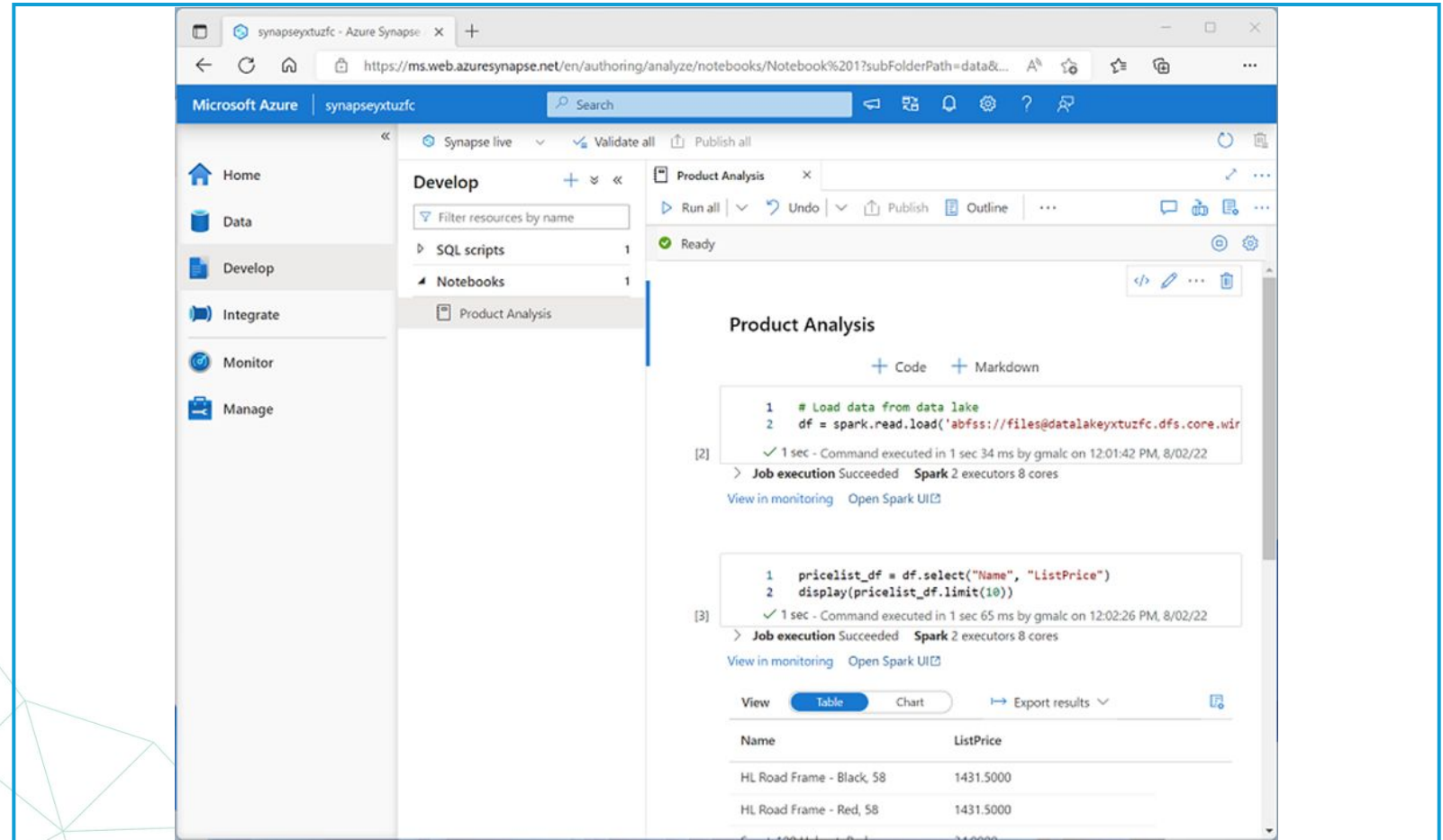
At the bottom of the Results pane, a status message indicates: '00:00:00 Query executed successfully.'

Process and analyze data with Apache Spark

Open-source Spark technology

- Highly scalable, distributed processing
- Common libraries and multiple programming languages

Integrated notebook experience



The screenshot displays the Microsoft Azure Synapse Studio interface. The left sidebar shows navigation options: Home, Data, Develop, Integrate, Monitor, and Manage. The 'Develop' section is active, showing a list of resources including 'SQL scripts' and 'Notebooks'. The 'Product Analysis' notebook is open, showing a code editor with Spark code and its execution results. The code includes loading data from a data lake and filtering it by name and list price. The execution results show a table with two columns: 'Name' and 'ListPrice'.

```
1 # Load data from data lake
2 df = spark.read.load('abfss://files@datalakeyxtuzfc.dfs.core.wir

[2] ✓ 1 sec - Command executed in 1 sec 34 ms by gmalc on 12:01:42 PM, 8/02/22
> Job execution Succeeded Spark 2 executors 8 cores
View in monitoring Open Spark UI

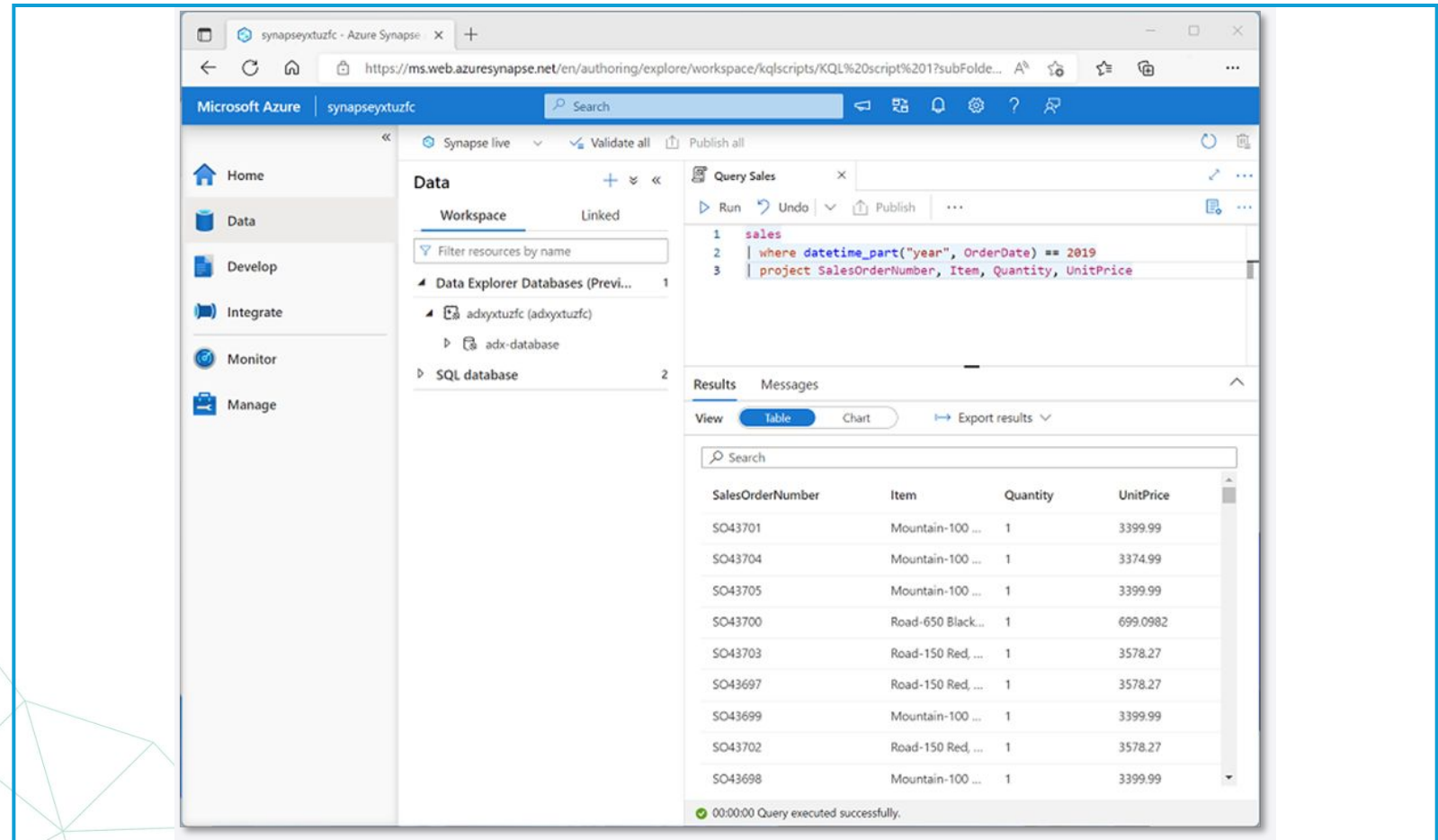
1 pricelist_df = df.select("Name", "ListPrice")
2 display(pricelist_df.limit(10))

[3] ✓ 1 sec - Command executed in 1 sec 65 ms by gmalc on 12:02:26 PM, 8/02/22
> Job execution Succeeded Spark 2 executors 8 cores
View in monitoring Open Spark UI
```

Name	ListPrice
HL Road Frame - Black, 58	1431.5000
HL Road Frame - Red, 58	1431.5000

Explore data with Data Explorer

- High-performance real-time data analytics
- Powerful, intuitive Kusto query language



The screenshot displays the Microsoft Azure Data Explorer web interface. The left sidebar shows navigation options: Home, Data, Develop, Integrate, Monitor, and Manage. The main area is divided into three sections: 'Data' (Workspace and Linked), 'Query Sales' (with a Kusto query editor), and 'Results' (with a table view). The query executed is:

```
1 sales
2 | where datetime_part("year", OrderDate) == 2019
3 | project SalesOrderNumber, Item, Quantity, UnitPrice
```

The results table shows the following data:

SalesOrderNumber	Item	Quantity	UnitPrice
SO43701	Mountain-100 ...	1	3399.99
SO43704	Mountain-100 ...	1	3374.99
SO43705	Mountain-100 ...	1	3399.99
SO43700	Road-650 Black...	1	699.0982
SO43703	Road-150 Red, ...	1	3578.27
SO43697	Road-150 Red, ...	1	3578.27
SO43699	Mountain-100 ...	1	3399.99
SO43702	Road-150 Red, ...	1	3578.27
SO43698	Mountain-100 ...	1	3399.99

A status message at the bottom indicates: "00:00:00 Query executed successfully."

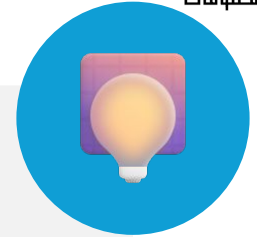
Exercise: Explore Azure Synapse Analytics

Use the hosted lab environment provided, or view the lab instructions at the link below:

<https://aka.ms/mslearn-explore-synapse>



Knowledge check



- 1 Which feature of Azure Synapse Analytics enables you to transfer data from one store to another and apply transformations to the data at scheduled intervals?
 - ☐ Serverless SQL pool
 - ☐ Apache Spark pool
 - ☒ Pipelines
- 2 You want to create a data warehouse in Azure Synapse Analytics in which the data is stored and queried in a relational data store. What kind of pool should you create?
 - ☐ Serverless SQL pool
 - ☒ Dedicated SQL pool
 - ☐ Apache Spark pool
- 3 A data analyst wants to analyze data by using Python code combined with text descriptions of the insights gained from the analysis. What should they use to perform the analysis?
 - ☒ A notebook connected to an Apache Spark pool
 - ☐ A SQL script connected to a serverless SQL pool
 - ☐ A KQL script connected to a Data Explorer pool

Learning Path Recap

In this learning path, we:

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Nunc viverra imperdiet enim. Fusce est. Vivamus a tellus.

Further reading

Get started with data engineering on Azure

<https://aka.ms/mslearn-data-engineer>

